Experimenting with algorithmic probability

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Problem: How can we measure algorithmic probability?

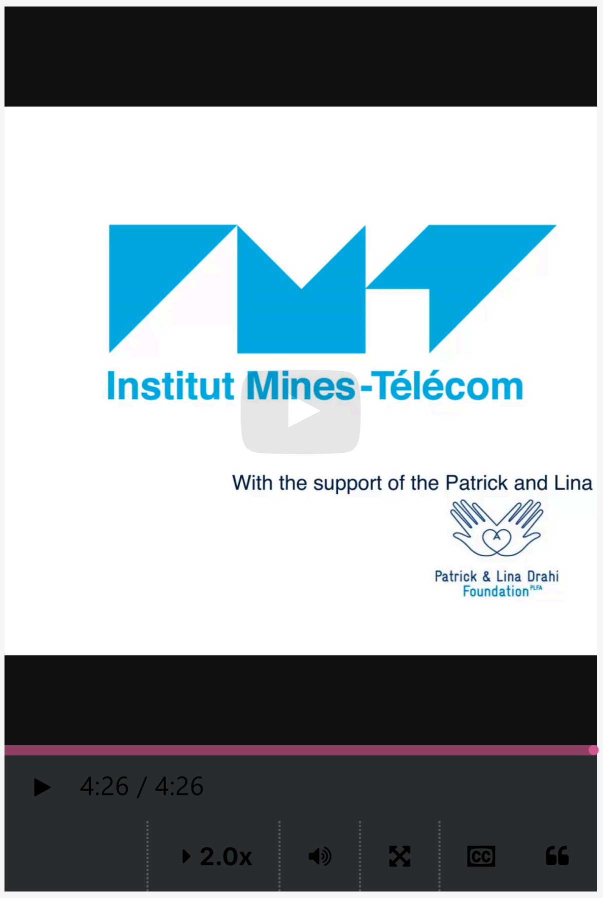
The main idea underlying algorithmic probability is that objects are constructed from scratch, and that the probability of their emergence depends on their complexity.

Question: Can we measure algorithmic probability by running random programs on a computer?

Answer: In a strict sense, no, as some of these programs may loop indefinitely. But in practice, yes, for simple machines.

Not only can algorithmic probability be approximated by trying random programs on simple machines, but this constitutes a direct way of approximating Kolmogorov complexity, since $K(s) \approx \log_2(1/P(s))$. This is the idea underlying The Online Algorithmic Complexity Calculator, developed by the Algorithmic Nature Group.

Estimating Complexity using Algorithmic Probability



programs

on all the possible 5-state Turing

machines!

Don't worry;

you won't have to run these programs

yourselves:

the results are available on the

Online Algorithmic Complexity Calculator

Website.

Let's have a look at it.

In the OACC experiment, binary strings are generated by simple Turing machines (five states only) (see section methodology). By sampling halting programs (within a maximum runtime) using

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